

BEHAVIOUR-ALIGNED FRONT-ENDS FOR FULL-DUPLEX SPEECH LLMs: SIGNAL RESTORATION IS NOT BEHAVIOUR RESTORATION

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ABSTRACT

Hosted full-duplex speech LLMs are absorbing signal clean-up, leaving on-device front-ends whose job is to serve the model’s *turn-taking decision*. We ask whether a suppressor trained on signal metrics serves that decision, and what changes when the same 78 k-parameter RNNoise-style front-end is aligned to the decision through a frozen model, scoring first-response placement on a noisy HumDial test set. On Lychee-FD (12.8 B), signal-trained suppression restores every clean-condition pathology, while behaviour-level training significantly cuts wrong-place responses, from 67 % to 48 %, without losing on-time responses, staying silent where noise hides the evidence. On Freeze-Omni (7B) a front-end has no leverage at moderate noise; under severe masking the frozen suppressor makes the model take over earlier and the decision-aligned one significantly later, with first-response relevance preserved. Code: <https://github.com/jianmliu/behaviour-aligned-frontends>.

Index Terms— full-duplex speech LLM, noise suppression, turn-taking, false barge-in, behaviour-aware front-end

1. INTRODUCTION

Full-duplex spoken dialogue models emit agent audio while ingesting user audio, so their core competence is a continuous decision: speak, keep listening, or back-channel. That decision is made from raw acoustics. For transcription-oriented pipelines, decades of work hardened recognition against noise; the *turn-taking* decision of a full-duplex model has no such history, and it fails differently—not by mis-hearing words but by *acting*: answering a slammed door, treating café babble as a barge-in.

The evidence is already on the record. The ICASSP HumDial challenge [1], whose dual-channel benchmark is the most recent standard for full-duplex interaction, reports that across submitted systems “robustness under background noise remains limited, with transient noises causing false activations or missed detections”—and yet the benchmark itself, like Full-Duplex-Bench and its successors [2, 3, 4], contains no noise condition. The gap reaches the commercial frontier: OpenAI’s GPT-Live-1, recently released to its API, is full-duplex, removes the separate turn detector from the audio path, reports a 30-point gain on Full-Duplex-Bench, and claims “improved

handling of ambient noise”—without a single noise-condition number [5]. The community has named the failure and does not yet measure it.

The deployed answer to noise is an upstream suppressor—in the open-source stack, most commonly RNNoise [6] and its descendants, built and tuned against signal metrics on data the downstream model never sees. Hosted full-duplex models change what such a front-end is *for*. Signal clean-up is being absorbed into the model; what stays on the device is what the cloud cannot do: decide, cheaply and privately, whether a sound is addressed to the system and whether a turn is coming—before a metered session sees it. A front-end with that job must be aligned to the *behaviour* of the model it serves. Whether signal-metric training does that, and what behaviour-level alignment changes, are the questions this paper measures, with the model held frozen throughout.

A companion line of work on *acoustic echo* found that even cross-entropy on the model’s own turn-taking tokens leaves an objective-behaviour gap dominated by teacher-forcing *exposure*: echo is the model’s own output fed back, and the states that break the benchmark are never visited under teacher forcing (companion work, in submission). Noise carries no such loop—it is exogenous, “do not activate on this” is an ordinary conditional decision fully covered by teacher-forced supervision. If behaviour-level training of a front-end works anywhere, it should work here.

Contributions. (1) *Signal restoration is not behaviour restoration*, on two models: the same 78 k front-end, trained on signal metrics, hands a frozen full-duplex model back its clean-condition pathologies (Lychee-FD) or makes it take over earlier (Freeze-Omni, at every DEMAND SNR); aligned to the model’s turn decision through gradients, it reverses that effect wherever the failure has an acoustic share, with leverage that tracks that share and no cost in first-response relevance. (2) The measurement instrument: a noise-augmented behavioural protocol on the HumDial test set scoring first-response *placement*, grounded in the benchmark’s respond/refrain semantics (§2). (3) On Lychee-FD the behaviour-trained front-end cuts the wrong-place total from 67 to 48 % ($p < 10^{-23}$, paired) at no cost in on-time responses (misses stay at the noise level), and ablations show the signal and behaviour solutions to be distinct attractors (§3). (4) Released protocol, metrics, recipes and logs.

2. SETUP

2.1. Model and behaviour semantics

Lychee-FD [7] (12.8B) emits explicit turn-taking *control tokens* every 0.4s chunk (keep listening, start speaking, backchannel, ...): its behaviour is directly observable and directly supervisable. Audio enters as continuous encoder features, so gradients pass through the frozen model into a front-end.

2.2. Noise protocol on HumDial

The HumDial English test set groups 4,550 recordings into ten scenarios with explicit respond/refrain semantics: four demand a response (ask, repeat, shift, deny); six demand restraint (backchannel, wait, pause, talk_to_others, two others_talk_to_user variants). We take 100/scenario, add noise from held-out DEMAND environments at $\text{SNR} \in [0, 15]$ dB (per-sample seeded: all systems see bit-identical scenes), and feed the frozen model offline. Offline decoding of these long, two-turn recordings yields, in practice, a single model response (93 % of clean decodes that respond), so we measure the *placement* of that response—a quantity with clean semantics per scenario and direct sensitivity to noise: **on-time** (at or after the end of the user’s question), **premature** (inside the question—a failure HumDial itself reports for current models on slow speech), **miss** (no response), **mis-addressed** (placed inside speech directed at someone other than the model: the third-party segments of the `others_*` and `talk_to_others` scenarios—the direct false-trigger read-out), and **break-preempt** (inside an annotated mid-question pause). Every comparison is a *paired* contrast on bit-identical samples, which is robust to an imperfect clean baseline: premature take-overs dominate it for both models we test, as HumDial’s organisers report for current systems, and offline decoding exposes them fully because nothing interrupts the model once it starts.

2.3. Front-end and training ladder

RN-t is a differentiable RNNoise-style suppressor: 22 Bark-band features, the original Dense-GRU-GRU-GRU-Dense gain topology (78k parameters), band gains interpolated to STFT bins, length-preserving. Signal-loss pretraining (SI-SNR, speech from a disjoint corpus, DEMAND ch01) reaches +9.3 dB held-out gain at SNR 5 with 32.5 dB clean-speech fidelity. **R-indep** freezes this checkpoint—deployment practice. **R-joint** fine-tunes it through the frozen model against cross-entropy on the model’s own clean-run control-token trajectory (self-distillation; the released transition-preserving label subsampling), on a training corpus disjoint from HumDial (Full-Duplex-Bench 1.0 user audio, 429 recordings ≤ 20 s, DEMAND noise re-drawn per step). The comparison isolates

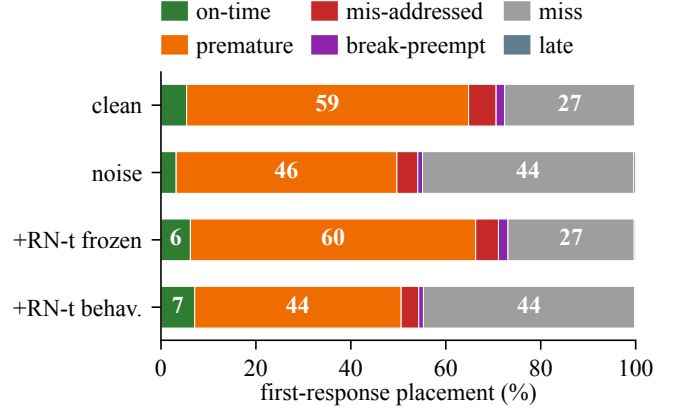


Fig. 1. First-response placement across the four conditions (1000 paired samples). Noise collapses responses into misses; the frozen suppressor restores the clean profile, pathologies included; the behaviour-fine-tuned one shrinks every wrong-place class instead.

Table 1. First-response placement on the HumDial test set (1000 samples, 100/scenario; identical noise per sample across all noisy conditions). Mis-addr. = response placed inside third-party speech (false trigger); break-pre. = response inside an annotated mid-question pause (% of the pause scenario). RN-t-frozen = signal-pretrained, frozen (R-indep); RN-t-behav. = same start, fine-tuned on control-token CE (R-joint).

%	clean	noise	+RN-t-frozen	+RN-t-behav.
on-time ↑	5.4	3.2	6.2	7.1
premature ↓	59.4	46.5	60.1	43.5
miss ↓	27.4	44.5	26.6	44.5
mis-addr. ↓	5.8	4.4	4.8	3.7
break-pre. ↓	18	10	20	10

exactly one variable: what the training objective knows about the model being served.

3. RESULTS

3.1. Noise breaks the turn-taking decision

Table 1, first two columns. The clean baseline already carries the pathologies HumDial reports for current systems—premature answers dominate (59.4 %; 86 % on `talk_to_others`), and 58 % of `others_talk_to_user_before` decodes fire inside speech addressed to a third party. Noise does not add new misbehaviour on top; it *defends*: misses rise in all ten scenarios (most on `ask`, 28 → 51), on-time responses halve, and every hearing-dependent failure—premature, mis-addressed, pause preemption—falls passively with the signal. The same direction reproduces on VoiceChat

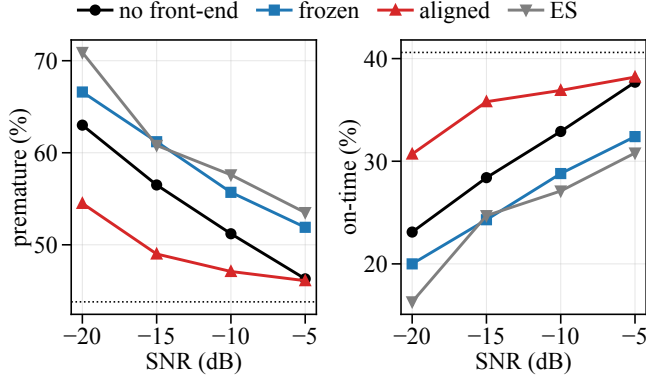


Fig. 2. Freeze-Omni, 1,000 paired samples per SNR: no front-end, frozen signal-trained suppressor, the same front-end aligned through gradients, and evolution strategies (ES) on decisions. Aligned is later than frozen at every SNR, later than none from -10 dB down. Dotted: clean.

11B under our companion echo protocol’s noise-only control (self-talk $1.60 \rightarrow 1.94$, premature take-overs $1.09 \rightarrow 1.39$ events/min).

3.2. Frozen versus behaviour-fine-tuned suppression

Table 1, last two columns, one training variable apart. The frozen suppressor is a faithful restorer: misses return to the clean level (26.6 %) and so does everything else—premature back to 60.1 %, pause preemption at 20 %, false triggers at 4.8 %. Signal fidelity hands the model back its clean-condition behaviour, pathologies included. The behaviour-fine-tuned front-end spends the same capacity differently: 17 pt fewer premature answers than the frozen variant (paired exact McNemar on bit-identical samples: 86 vs 252 discordant pairs, $p = 5 \times 10^{-20}$), the lowest false-trigger rate of any condition (3.7 %), and half the pause preemptions—while misses stay at the noise level (44.5 %, $p = 2 \times 10^{-22}$). Aggregating the three wrong-place classes (premature + mis-addressed + break-preempt) gives 66.9 % for the frozen front-end against 48.2 % for the behaviour-trained one ($p = 4 \times 10^{-24}$), at no measurable cost in correct responses: on-time is 7.1 % against the frozen variant’s 6.2 % and the clean baseline’s 5.4 % (both n.s.). The trade it learns is legible: speak when the decision evidence survives the noise, stay silent when it does not.

3.3. Second model: leverage tracks the acoustic share

To test whether the Lychee-FD finding is specific to one model, we repeat the protocol on Freeze-Omni [8] (7B, frozen Qwen2 backbone, a chunk-level *state head* that decides when to speak; listening needs no token generation, so placement is read from that decision alone) on the same 1,000-sample protocol (200 samples, 20/scenario, for the self-distillation variant and the clean-audio and probe rows), and train the same 78 k front-end

from the same signal-pretrained start (Full-Duplex-Bench audio, DEMAND noise drawn around the evaluation SNR) with a differentiable *expected placement cost* (first-trigger distribution over chunks $\times$ premature/on-time/miss costs, refrain samples penalising any trigger, plus a logit-margin term in the premature region so that a saturated decision still carries gradient) through the frozen model. Freeze-Omni’s clean profile is the mirror image of Lychee-FD’s: on-time 40.6 %, premature 43.8 %, no misses—and no restraint: it fires inside third-party speech in 99/100 *others_before* samples and inside the pause gap in 57/100. Noise at the protocol’s SNR (0–15 dB) does not move it at all (on-time 40.6 %, premature 43.4 %, restraint axes 100/100 and 60/100): an eager model is not deafened. Pushed further, an acoustic share does appear—but as *more* eagerness, not silence: from -5 to -20 dB premature take-overs rise from 46.3 to 63.0 % (Fig. 2; 1,000 samples each) with still no misses; sustained masking acts as a trigger, whereas 300 ms bursts at $+10/+20$ dB barely register (42.5/40.0, 39.5/43.5). Self-distillation cannot be the target (the model’s own clean first response precedes the annotated turn end by a median 7.3 s), so we use the benchmark’s turn-end annotations; on clean audio the front-ends leave placement unchanged (200 samples, n.s.). Under noise, the frozen suppressor nudges the model slightly *more* eager (on-time 38.8 %, premature 45.1 %, restraint axes 100/100 and 61/100: cleaner speech, if anything earlier take-over). The behaviour-trained front-ends land on the noisy baseline—soft placement with annotation targets 39.7/44.3 on 1,000 samples, restraint axes 100/100 and 60/100 exactly as without it; on 200 samples, self-distillation targets 43.0/39.0 (on-time/premature), all within sampling noise. Per sample, the front-ends move decision instants the *wrong* way for restraint—earlier take-overs outnumber later ones 172:115 (frozen) and 156:91 (trained), sign test $p < 10^{-3}$: cleaner speech makes an eager model marginally more eager. The trained suppressor kept 6.4 of the frozen model’s 8.6 dB SI-SNR gain on Full-Duplex-Bench mixtures: nothing collapsed, and the only direction found was the acoustic one. *Where the acoustic share exists, the leverage appears.* At -15 dB (1,000 paired samples; 28.4/56.5 without a front-end) the frozen suppressor makes matters worse—24.3/61.2, decisions earlier 340:174 (sign test $p = 2 \times 10^{-13}$): denoising feeds the trigger. The behaviour-trained front-end moves them the other way: 35.8/49.0, later 309:173 against no front-end ($p = 6 \times 10^{-10}$) and 432:139 against the frozen suppressor ($p < 10^{-35}$). Training at the evaluation SNR does not explain this: the same suppressor retrained on SI-SNR with the same data, noise and steps restores the signal best (+18.4 dB SI-SNR on these mixtures vs. +17.1 frozen and +12.5 aligned) yet still takes over earlier than no front-end (25.8/59.1; 281:213, $p = 0.003$) and far earlier than the aligned one (357:159, $p < 10^{-17}$). Capacity is the other route: DeepFilterNet3 (2.1 M parameters, +21.5 dB) takes over later than no front-end (33.2/51.6; 323:262, $p = 0.01$) and is indistinguishable from the aligned 78 k front-end (262:299, $p = 0.13$). At 78 k

only the decision objective reaches that placement. The same 78 k parameters, aligned to the decision instead of the signal, reverse the front-end’s effect: the Lychee-FD finding on a second model. A hosted API would expose only the decisions; in a small probe, evolution strategies on the realised placement reward (23 band gains or all 78 k parameters, up to 9k rollouts, also with the gradient arm’s loss as reward) recovered at most a small part of this effect (Fig. 2). The pattern holds at -10 and -20 dB (sign tests $p < 10^{-5}$); at -5 dB alignment only cancels the frozen suppressor’s harm (Fig. 2). Content follows: judged by Qwen2-7B-Instruct on the opening of the first response (0–2 relevance, -15 dB), the frozen suppressor slightly lowers relevance (0.77 vs. 0.83, n.s.), gradient alignment keeps it (0.84; 170:132 above frozen, $p = 0.03$) and cuts hallucinated Chinese openings from 13.6 to 9.4 % ($p = 2 \times 10^{-4}$); later take-overs cost no relevance. A second, unseen corpus (MUSAN noise) repeats the structure at -15 dB: no help from the frozen suppressor (14.2/74.3 vs. 13.0/75.6, $p = 0.24$), no significant transfer of the DEMAND-trained 78 k front-end (15.8/71.2, $p = 0.22$), a clear effect once retrained on MUSAN (275:151 later than no front-end, $p = 2 \times 10^{-9}$): the recipe transfers, the weights do not. Retrained on MUSAN with the signal loss instead, the suppressor gains far more SI-SNR (+9.5 dB vs. +1.5 frozen and +0.8 aligned) yet takes over even earlier than no front-end (9.8/78.5; 404:282, $p < 10^{-5}$). Across regimes, a waveform front-end has leverage only over the acoustic share of a failure—large on Lychee-FD, zero here at the protocol’s SNR and growing with masking.

3.4. Ablations: two attractors, no middle ground

Three variants probe what the behaviour objective needs. (i) *Gentler learning rate* (3×10^{-5} vs 10^{-4}): the fine-tune never leaves the pretrained signal solution—its profile is indistinguishable from the frozen front-end (on-time 6.2 %, premature 62 %, miss 25 %). (ii) *Scratch start*: without signal pretraining the same objective learns nothing usable; the profile matches raw noise (on-time 3.6 % vs 3.2 %). (iii) *SI-SNR anchor* added to the CE ($\lambda=0.05$): the anchor dominates and the solution reverts to the frozen profile, restraint gains included. The picture is consistent: the signal solution and the behaviour solution are distinct attractors—behaviour training must start from signal competence (ii), needs enough learning rate to escape it (i), and is not reachable as a regularised compromise (iii). The objective is not a seasoning over signal fidelity; it selects a different front-end.

4. DISCUSSION

Where front-ends still pay. Egocentric-device challenges chart the decline of the generic front-end: in CHiME-8 MM-CSG [9] training data beat array processing on WER, yet a neural separation front-end won the *speaker-attribution* axis [10]; by SLT 2026 SmartGlasses [11] the winner used no front-end

at all. The opposite reaction grows the front-end into a 30B LLM trained on task labels (UAF [12]). Our results locate the surviving value: a front-end pays on the axis its objective shares with the downstream decision—attribution there, turn-taking here—and 78 k parameters aligned to that decision reach the placement of a $27\times$ larger signal restorer, which signal training of the same 78 k does not. Deployment follows: with hosted full-duplex models metered per minute of audio, every noise-induced wrong-place response is a billed event, and an on-device front-end that reduces them at zero marginal cost is a cost control, not an audio-quality option. Unlike enhancement-for-ASR joint training, the target is behaviour, not transcription.

Relation to the echo domain. In the companion echo setting the same CE failed to move free-running behaviour because teacher forcing never visits the states echo creates; here the perturbation is exogenous and the same objective moves behaviour—consistent with the exposure account. What remains is an *objective* gap: the CE optimum under noise is a restraint profile, reachable only from a signal-competent start (§3.4).

Limitations. English subset only; offline decoding of the official test set (no streaming latency axis); front-ends trained through two models (the VoiceChat corroboration is measurement-only); two noise corpora (DEMAND, MUSAN), each held out between pretraining and test, and a front-end must be trained on its target noise; one LLM judge, opening sentences only; matched-SNR controls on Freeze-Omni only, DeepFilterNet3 on DEMAND only; the gradient-free probe is zeroth-order search at small budgets on one model.

5. CONCLUSION

We measured what noise does to the turn-taking decision of a frozen full-duplex speech LLM on the benchmark that first named the problem, and compared the deployed remedy—a signal-trained suppressor—against the same suppressor fine-tuned on the model’s own turn-taking tokens. Signal training restores what the model hears; behaviour training changes what the model does: the wrong-place total falls from 67 to 48 % ($p < 10^{-23}$, paired), on-time responses are preserved, and the cost is silence where the evidence does not survive the noise. The two solutions are separate attractors, not ends of a dial. On a second, eager model, behaviour alignment reverses the frozen suppressor’s effect once masking creates an acoustic failure. The choice is not how much to denoise, but which decision the denoiser should serve.

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